Math 257 Definitions and Theorems

Week 8: Modules 26-29

Fall 2023

26 Determinants

Goal: Define a determinant function that takes an $n \times n$ matrix A to a real number $\det(A)$, such that $\det(A) \neq 0$ if and only if A is invertible.

We start with small matrices:

Definition 26.1. The **determinant** of

- a 1×1 matrix is $\det([a]) = a$.
- a 2×2 matrix is $\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$.

Notation. We will write both $\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right)$ and $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ for the determinant.

Definition 26.2. The **determinant** is the function that assigns to each $n \times n$-matrix a number and satisfies the following conditions:

- **(Normalization)** $\det I_n = 1$,
- It is affected by elementary row operations as follows:
  - **(Replacement)** Adding a multiple of one row to another row *does not change* the determinant.
  - **(Interchange)** Interchanging two different rows *reverses* the sign of the determinant.
  - **(Scaling)** Multiplying all entries in a row by s, *multiplies* the determinant by s.

Theorem 26.1. *The determinant of a triangular matrix is the product of the diagonal entries.*

Theorem 26.2. *Let A be an $n \times n$-matrix, then $\det(A) = 0$ if and only if A is not invertible.*

Theorem 26.3. *Let A, B be two $n \times n$-matrices. Then $\det(AB) = \det(A)\det(B)$.*

Theorem 26.4. *Let A be an $n \times n$-matrix. Then $\det(A^T) = \det(A)$.*

27 Cofactor expansion

Notation. Let A be an $n \times n$-matrix. We denote by A_{ij} the matrix obtained from matrix A by deleting the i-th row and j-th column of A.

Definition 27.1. Let A be an $n \times n$-matrix. The $(\mathbf{i}, \mathbf{j})$**-cofactor** of A is the scalar C_{ij} defined by

$$C_{ij} = (-1)^{i+j} \det A_{ij}.$$

Theorem 27.1. *Let A be an $n \times n$-matrix. Then for every $i, j \in \{1, \ldots, n\}$*

$$\begin{aligned} \det A &= a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} \qquad (\textit{expansion across row } i) \\ &= a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj} \qquad (\textit{expansion down column } j) \end{aligned}$$

28 Eigenvectors

Definition 28.1. Let A be an $n \times n$ matrix. An **eigenvector** of A is a *nonzero* $\mathbf{v} \in \mathbb{R}^n$ such that

$$A\mathbf{v} = \lambda\mathbf{v},$$

where λ is a scalar, known as the **eigenvalue** associated with $\mathbf{v}$.

Definition 28.2. Let λ be an eigenvalue of A. The **eigenspace** of A associated with λ is the set

$$\mathrm{Eig}_\lambda(A) := \{\mathbf{v} \ : \ A\mathbf{v} = \lambda\mathbf{v}\}.$$

It consists of all the eigenvectors of A with eigenvalue λ and the zero vector.

29 Computing Eigenvalues and Eigenvectors

Theorem 29.1. *Let A be an $n \times n$ matrix and λ be a scalar. Then λ is an eigenvalue of A if and only if* $\det(A - \lambda I) = 0$.

Theorem 29.2. *Let A be an $n \times n$ matrix. Then $p_A(t) := \det(A - tI)$ is a polynomial of degree n. Thus A has at most n eigenvalues.*

Definition 29.1. We call $p_A(t)$ the **characteristic polynomial** of A.

Theorem 29.3. *Let A be $n \times n$ matrix and let λ be eigenvalue of A. Then*

$$Eig_\lambda(A) = Nul(A - \lambda I).$$